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DEVELOPMENT AND EVALUATION OF A SODIUM ALGINATE AND HPMC-BASED IN SITU GELLING GASTRORETENTIVE SYSTEM OF CURCUMIN FOR ANTI-ULCER THERAPY

Subashis Sahoo - Research Scholar

Department of Pharmacy, Kalinga University, Raipur, Chhattisgarh

Ayushman Roy - Assistant Professor

Faculty of Pharmacy, Kalinga University, Raipur, Chhattisgarh

ABSTRACT:

The use of curcumin for peptic ulcers is deeply hampered due to its poor water solubility, fast systemic metabolism and low oral bioavailability. To overcome these drawbacks, the present study was aimed to design, formulate and evaluate gastroretentive drug delivery system of curcumin which is a floating in-situ gelling system to increase its gastric residence time and efficacy. The formulations (F1–F9) were developed by utilizing the ionic gelation method by varying the concentration of sodium alginate as the main ion-sensitive gelling agent and HPMC K4M as structural polymer for viscosity. Calcium carbonate was used for triggered cross-linking and sodium citrate was used to delay gelation. Preformulation study by FTIR and DSC showed total drug-excipient interaction. When placed in the simulated gastric fluid (0.1 N HCl, pH 1.2), the formulations were able to sol-gel rapidly forming strong, highly buoyant matrices. The rheological properties, mechanical strength of the gel and floating of the optimized batches (F8 and F9) which contained more polymer concentration were better. In vitro dissolution studies revealed sustained curcumin release over a period of 12 hours. Using kinetic mathematical modelling, the mechanism of drug diffusion could best be confirmed by the Higuchi model that followed Korsmeyer-Peppas kinetics which was non-Fickian (anomalous). This floating in-situ gel platform can optimize the delivery of curcumin for gastric mucosal disorders in a very effective, targeted manner.

Keywords: *Curcumin; Floating in-situ gel; Gastroretentive drug delivery system (GRDDS); Sodium alginate; Peptic ulcer; Sustained release; Ionic gelation; HPMC K4M.*

Introduction

According to WHO, peptic ulcer disease (PUD) presents as common complex gastrointestinal disorder worldwide with deep lesions of mucosa stomach and duodenum. The basic pathogenesis of peptic ulcer disease (PUD) is an important imbalance between the aggressive endogenous factors like the gastric acid and the defensive mucosal factors. Infection with Helicobacter pylori, using alcohol and prolonged use of non-steroidal anti-inflammatory drugs (NSAIDs) (Lanas & Chan, 2017; Kevitt et al., 2019) amplify the imbalance further. Commonly used treatment methods rely greatly on the usage of either proton pump inhibitors (PPIs) or histamine H2-receptor. But conventional oral dosage forms often get emptied out from the stomach quickly, which substantially limits their stay in the stomach. This bodily limit causes which leads to incomplete absorption of drug, the requirement of frequent doses, and likely systemic side effects, therefore limits their therapeutic efficacy at the ulcer site (Mandal et al., 2016; Pawar et al., 2011). There is thus an essential clinical need for the development of novel drug delivery systems that can prolong the gastric retention of therapeutics to achieve high localized concentrations and enhance patient compliance (Pawar et al., 2011).

Curcumin, a naturally occurring polyphenol that originated in the rhizomes of Curcuma longa (turmeric), has received much interest for its pleiotropic properties (Heger et al., 2014). It is known to be powerfully anti-inflammatory, antioxidant, antimicrobial, and tissue-healing properties (Kwiecien et al., 2019; Mei et al., 2012). Curcumin has been shown to decrease inflammatory cytokines, neutralize reactive oxygen species, inhibit gastric acid secretion, and regenerate the mucosal lining (Kwiecien et al., 2019; Mei et al., 2012). Although curcumin is potent, its clinical potential is limited. It is aqueous form, but it is soluble in water, is rapidly metabolized, and oral bioavailability is extremely low (Anand et al., 2007). To capitalize on the localized healing properties of curcumin in gastric ulcers, it is important to obtain curcumin into a form that avoids these limitations and is absorbed directly into the gastric mucosal lining.

Gastroretentive drug delivery systems (GRDDS) address the shortcomings of oral agents by extending the time that the drug is in the stomach (Tripathi et al., 2019). Floating in-situ gelling systems have proven to be very promising amongst the other GRDDS technologies. These systems are in the form of low viscosity, free-flowing liquid. The polymer matrix transforms into gel when it comes across the acidic conditions of the stomach (pH 1.2) and in a quick, pH-responsive fashion upon ingestion and exposure to the acidic habitat of the stomach (Rajinikanth et al., 2007). The gel becomes buoyant and floats on the stomach through the retention of carbon dioxide locally generated or the low density of the cross-linked polymer network. This

allows the system to release the drug in a controlled manner without the stomach's moving motor complex (Nayak et al., 2010).

We depend on the appropriate selection of polymers to create this gelling system. Biodegradable polymers are the best since they are biocompatible, low toxic and safe polymers (George & Abraham, 2006). Gelatine is a naturally occurring proteinaceous polymer, which is a by-product of collagen that has partially undergone hydrolysis and has excellent film-forming, mucoadhesive and structure-forming properties (El Zoghby, 2013). Adding gelatine into an in-situ gelling structure can be helpful alongside other pH-sensitive or ion-triggered natural polysaccharides to enhance the rheological characteristics, mechanical strength and bio-adhesion of the gel (Zhu et al., 2015). This poly (methyl acrylate)-poly (lactic acid) blend makes the gel resistant to gastric peristalsis and alters the release kinetics of drugs.

Materials and Methods:

Materials:

Curcumin was the main pharmaceutical ingredient. Sodium alginate and Hydroxypropyl methylcellulose (HPMC K4M) were used as gel-forming and viscosity-changing polymers (Zatz& Woodford, 1987). Calcium carbonate and calcium chloride were used as ionic cross-linking and gas-forming agents, and sodium citrate was used as a chelating agent to prevent premature gelation (Miyazaki et al., 2000). Methylparaben and propylparaben were used as antimicrobial preservatives. All chemical reagents and solvents used in this study were analytical grade.

Preformulation and Compatibility Studies

Curcumin measured at a wavelength of absorbance maxima λmax of 406 nm using a UV-Visible spectrophotometer in 0.1 N HCl (pH 1.2) (Kakran et al., 2012). To be processed optimally, the micromeritic properties of the powders were evaluated including bulk density, tapped density, angle of repose, Carr's index, and Hausner's ratio (Aulton & Taylor, 2013). To test drug-excipient compatibility, we stored 1:1 physical mixture of curcumin, sodium alginate, HPMC K4M at accelerated temperatures (40°C 2°C / 75% 5% RH) for 1 month at 40°C 2°C / 75% 5% RH). FTIR and DSC confirmed that there were no significant chemical interactions or structural changes (Mura, 2015; Verma & Iyer, 2000).

Preparation of the *In-Situ* Gelling System

The liquid formulation batches (F1–F9) were prepared as ionic gel. Sodium alginate (1.25–1.75% w/v) was first dissolved in distilled water at 60°C followed by the introduction of HPMC K4M (0.25–0.75% w/v) through magnetic stirring (Itoh et al., 2006). We used sodium citrate (0.25% w/v) to lock calcium ions in the solution after the solution cooled below 40°C. Following this step, we added calcium carbonate (0.5%–1.0%), calcium chloride (0.30% w/v) and parabens, all combined (Patel et al., 2011). Finally, we added dropwise a concentrated solution of curcumin (2.0% w/v) dropwise. We final volume was set at 100 mL and pH neutralization was maintained sol state (Rajinikanth & Mishra, 2008).

Table 1: Formulation Composition (Raw Materials % w/v)

Component	F1	F2	F3	F4	F5	F6	F7	F8	F9
Sodium Alginate	1.25	1.25	1.25	1.50	1.50	1.50	1.75	1.75	1.75
HPMC K4M	0.25	0.50	0.75	0.25	0.50	0.75	0.25	0.50	0.75
Calcium Carbonate	0.50	0.75	1.00	0.50	0.75	1.00	0.50	0.75	1.00
Sodium Citrate	0.25	0.25	0.25	0.25	0.25	0.25	0.25	0.25	0.25
Calcium Chloride	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30
Methylparaben	0.20	0.20	0.20	0.20	0.20	0.20	0.20	0.20	0.20
Propylparaben	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05
Curcumin	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
Distilled Water	QS	QS	QS	QS	QS	QS	QS	QS	QS

(Note: QS = Quantity Sufficient to reach 100 mL)

Physicochemical Evaluation and *In Vitro* Buoyancy:

The formulations of the liquids were analysed in terms of clarity, pH, total drug, and viscosity (determined with a Brookfield Digital Viscometer at 37°C) (Peppas et al., 2000). To determine the phase transition and the physical strength of the formulations, 1mL of each formulation was immersed in simulated gastric fluid (0.1N HCI, pH 1.2). Gelation times, matrix swelling indices (over 5 hours), and mechanical strengths of the gels were measured. The gastric retention properties of the gel were evaluated by a USP Type II apparatus (50 rpm, 37 C); the floating time and lag time were measured to assess the gastroretentive properties of the gel (Baumgartner et al., 2000).

***In Vitro* Release Profiling and Kinetic Modelling:**

To determine in vitro dissolution, a USP Type II apparatus that had 900 mL of 0.1 N HCl (pH 1.2) was used at 37 o C with paddle speed of 50 rpm. Aliquots were collected 6 hourly with cumulative release being quantified at 406 nm (Higuchi, 1963). To further explore the mass transport mechanisms whereby the curcumin was released out of cells, in vitro release values were modeled against Zero-Order, First-Order, Higuchi and Korsmeyer-Peppas models (Korsmeyer et al., 1983).

Results and Discussion:

5. Preformulation Study:

5.1. Physicochemical properties:

5.1.1 Organoleptic properties:

Organoleptic assessment of curcumin was conducted to assess the basic physical properties of curcumin, which is essential in identification and initial quality measurements. Curcumin was found to be a bright yellow to orange crystal with distinctive odor and slightly bitter flavor. The sample was homogeneous and fine which signifies uniformity and lack of impurities which can be seen. The identified and purified curcumin was reportedly known to exhibit these properties, thus confirming its identity and purity at an initial level.

Table 2: Organoleptic properties:

Physical Property	Standard	Sample
Colour	Yellow to Orange Crystalline Powder	Yellow to Orange Crystalline powder
Odor	Odourless or faint characteristic Odor	Odourless
Taste	Slightly bitter	Slightly bitter

5.1.2 Melting Point:

The capillary method was used to determine the melting point of curcumin to be 183.66 °C. High purity of the sample is reflected in the narrow melting range. But there were slight softening and gradual degradation towards the melting range, which is suggestive of limited thermal stability and is characteristic of curcumin. The identification of the drug and the crystallinity of the sample are indicated by the melting point value, which is consistent with literature values. This range would be considerably affected if there were contaminants or degradation products. As a result, determining the melting point also indicates its formulation suitability.

Table 3: Melting Point

Name of API	Practical Value	Standard value
	183.65ºC	
Curcumin	183.25ºC	183 - 186ºc
	184.10ºC	

5.1.3 Solubility Properties:

To learn how curcumin would act in the process of formulation, its solubility in different solvents was compared. This was found to be because curcumin is virtually insoluble in water, with extremely low aqueous solubility (under 0.0004 mg/mL at room temperature). A significant weakness that has an impact on its dissolution rate and bioavailability is its poor solubility. Conversely, curcumin demonstrated excellent dissolution in organic solvents like ethanol, methanol, acetone and dimethyl sulfoxide (DMSO). It was also found that the drug became more soluble in alkaline condition and this may be attributed to the ionization of its phenolic hydroxyl groups. As it was observed though, in acidic PH degradation of curcumin will occur

meaning that solubility will increase but the chemical stability can be affected. These findings clearly show that curcumin is hydrophobic and lipophilic in its nature and could not be readily put into aqueous systems. To address these constraints thus new drug delivery methodology, like in-situ gels, nanoparticles or solubilization should be used to increase its solubility and bioavailability.

Table 4: Solubility Properties:

Medium / Solvent	Solubility of Curcumin
Water	Practically insoluble
Ethanol	Soluble
Methanol	Freely soluble
Acetone	Freely soluble
Dimethyl Sulfoxide (DMSO)	Highly soluble
Chloroform	Soluble
Ethyl Acetate	Soluble
Alkaline solution (pH > 7)	Soluble (due to ionization)
Acidic buffer (pH 1.2)	Very slightly soluble
Phosphate buffer (pH 6.8)	Poorly soluble

5.1.4Analytical Characterizations:

5.1.4.1 Detection of a Drug using Differential Scanning Calorimetry (DSC):

The researchers used DSC to analyze the thermal properties and identity and purity assessment of the base curcumin material. The DSC thermogram (Figure 1) displayed a distinct endothermic peak which occurred at 173.85 °C and represented the melting point of curcumin. The melt point for compendial purposes in the United States Pharmacopeia (USP) monograph has been experimentally established to fall between 172 and 176 °C. The peak displays a sharp characteristic which lacks any secondary endothermic or exothermic peaks thus establishing that curcumin exists in its pure crystalline form without any thermal degradation or contamination. The thermogravimetric measurements verified the authentic status of curcumin from the source material which maintained solid pre-formulation stability and met pharmaceutical grade purity standards.

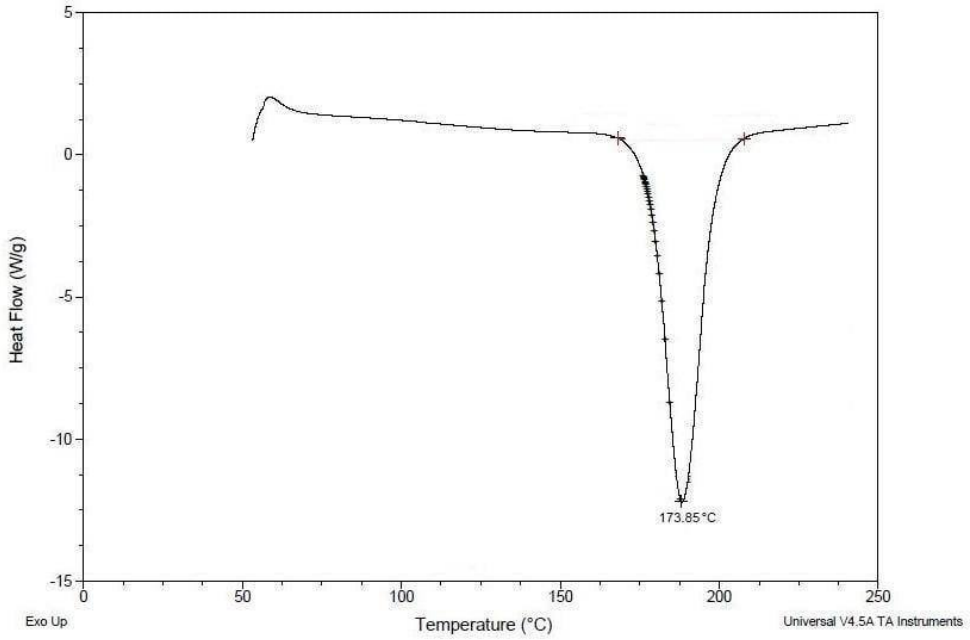


Fig. 15: DSC Study of Curcumin

5.1.4.2 Drug Identification Using FTIR Spectroscopy:

The active pharmaceutical ingredient's structural identity and functional group composition were confirmed through FTIR spectroscopy analysis. The FTIR spectrum showed a complete resolution which confirmed the identity of pure curcumin (Figure 2). The spectral data identifies phenolic hydroxyl groups (Broadband at 3505 cm1), conjugated double bonds (3277 cm1) and benzene-type aromatic rings (1627 cm1 and 1600 cm1), and the signature conjugated -diketone moiety (Intense peak at 1531 cm1, mixed C=O and C=C stretching). The spectrum presents absorption bands which support the structure by showing olefinic C–H bending at 1426 cm1 and phenolic ether linkages at 1276 cm1 and ether stretching at 1150 cm1. The spectrum displayed no abnormalities which may be associated with degradation products or impurities; the frequencies spectra coincided with the British Pharmacopoeia standard, confirming the chemical identity and purity of this drug.

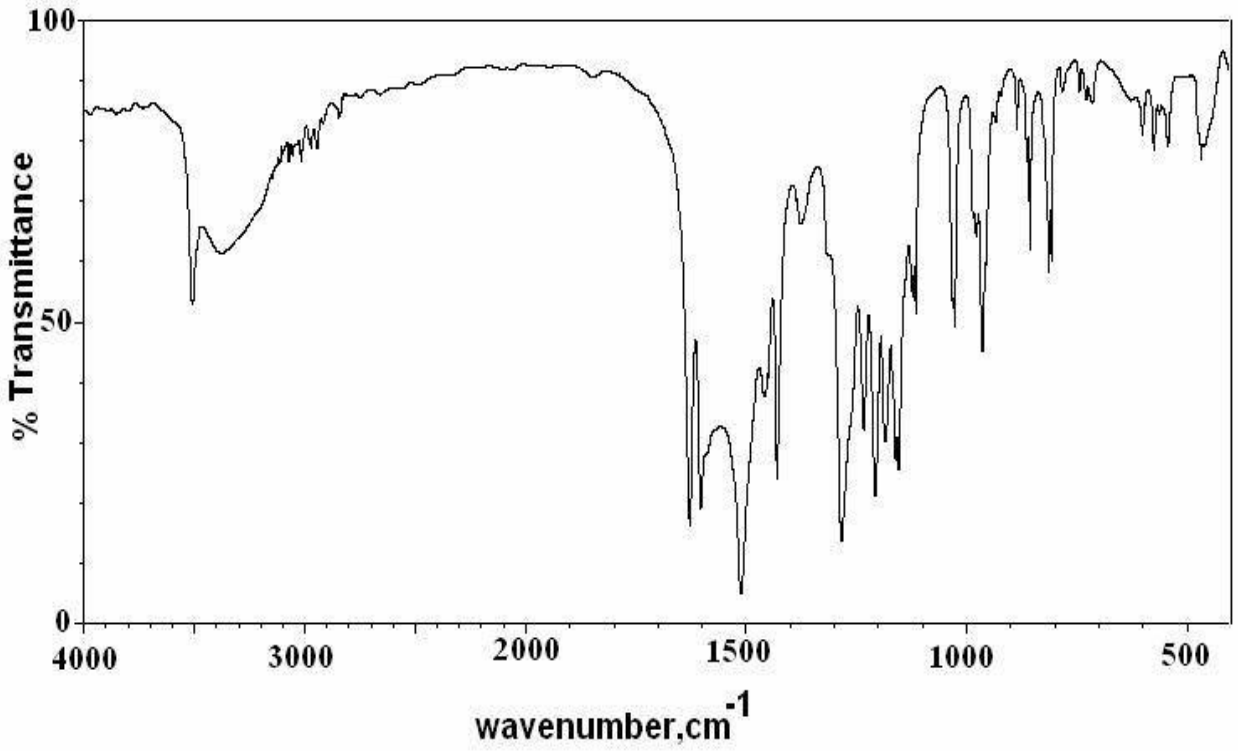


Fig. 16: FTIR Study of Curcumin

5.1.5 UV Absorption Maximum (λmax) Measurement:

To confirm the identity and provide a basis for quantitative analysis, we UV-Visible spectrophotometric analysis of the curcumin sample in 0.1 N HCl (pH 1.2), in the manner of human gastric fluid, showing a peak of maximum absorption at 406 nm (lambda_max). This peak corresponds to thepi right arrow pi* electronic transition that reveals the extensive conjugated network of aromatic rings, diketone, and double bonding system, which gives curcumin its yellow coloration. The presence of this sharp peak and strong peak without bands confirms the chemical purity of the sample and the absence of other contamination. Since this is the experiment indicates an exact relationship between lambda_max and published pharmacological literature, 425 nm was used as the standard wavelength for subsequent quantitative processes, calibration curves, and in vitro drug release.

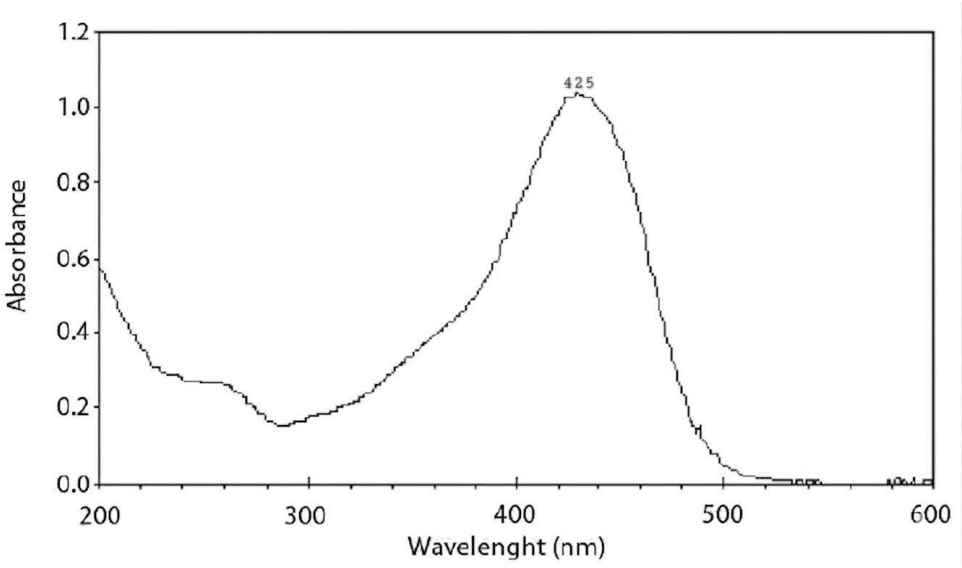


Fig. 17: λmax determination of Curcumin

5.1.5 Standard Calibration Curve of Curcumin:

To use UV-visible spectrophotometry to quantify curcumin in a simulated stomach (0.1 N HCl pH 1.2) at an absorbance maximum of 406 nm, a calibration curve was generated. The samples' absorbances were then measured, and a highly linear relationship was observed between the concentration (10-50 µg/mL) and the absorbances of the serially diluted standard reference solutions. The drug obeyed the Beer-Lambert Law at these concentrations with the almost perfect correlation coefficient (R^2 0.99) because of this linear regression analysis. This demonstrates the spectrophotometric method's accuracy and reproducibility, giving a linear regression equation.

(A = mC + c) can be effectively used to determine unknown curcumin concentrations in future in-vitro dissolution and total drug content analyses. Below: Absorbance data

Table 5: Absorbance Data for the Curcumin Calibration Curve:

Concentration (µg/mL)	Absorbance
10	0.189

20	0.368
30	0.599
40	0.798
50	0.956

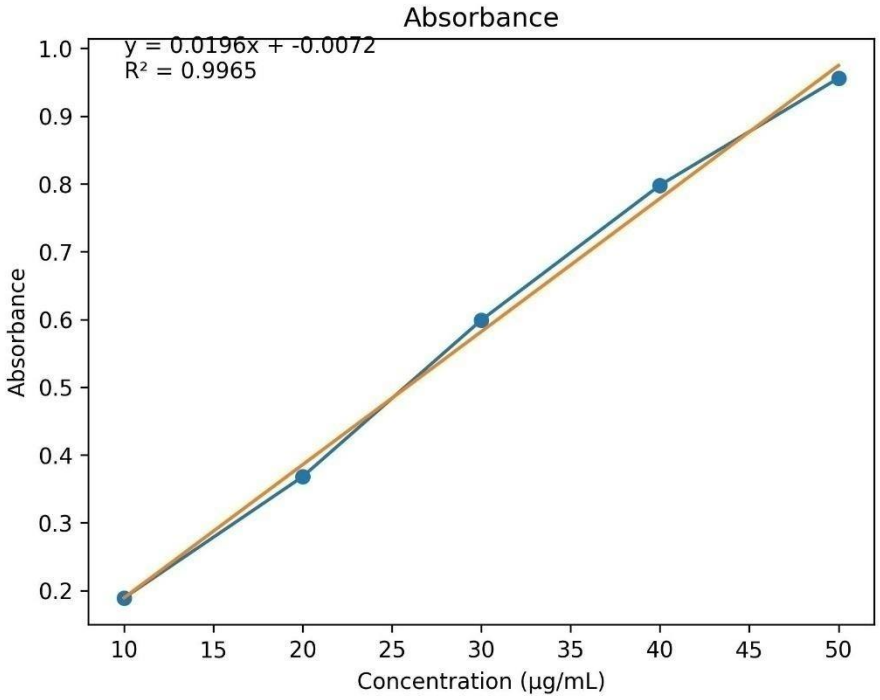


Fig.18: Calibration Curve of Curcumin

C) Drug Compatibility Studies :

1. Drug Compatibility Studies : (DSC)

Drug–polymer compatibility evaluations is a critical components of preformulation studies to evaluate possible physical and chemical interactions between the API and selected excipients. Such interactions may influence the stability, efficacy, and release behavior of the drug in the final dosage form. Therefore,

compatibility assessment plays a crucial role in ensuring the development of a stable and effective pharmaceutical formulation.

Table 6: DSC Spectra Studies

S. No.	Drug + Polymer Combination	Ratio	Storage Condition	Observation (2nd Week)	Observation (4th Week)	DSC Interpretation
1	API + Sodium Alginate	1:1	40°C ± 2°C / 75% RH ± 5% RH	No physical change	No physical change	No shift in melting peak
2	API + HPMC K4M	1:1	40°C ± 2°C / 75% RH ± 5% RH	No physical change	No physical change	Melting peak retained
3	API + Sodium Alginate + HPMC K4M	1:1:1	40°C ± 2°C / 75% RH ± 5% RH	No colour change	No significant change	No additional peaks observed

The study assessed how well the drug can work with sodium alginate and HPMC K4M polymer materials. The API was mixed with each polymer in a 1:1 ratio using precisely measured amounts of both substances. The mixtures were blended completely with a mortar and pestle until they reached uniformity before they were sifted through a #40 sieve for proper mixing. The prepared samples were transferred into glass vials which they sealed with rubber stoppers and aluminum caps before they stored the samples under accelerated stability conditions at 40°C ± 2°C and 75% ± 5% relative humidity for a month of period.

The researchers assessed physical properties of the samples through regular checks which included color assessment and odor detection and texture analysis to find any visible changes. The researchers utilized Differential Scanning Calorimetry (DSC) to assess how well the prepared mixtures would work together. The samples underwent scanning from 50°C to 220°C while a nitrogen atmosphere protected them from oxidative degradation. The resulting thermograms were examined to compare the melting behaviour of the pure drug with that of the drug–polymer mixtures.

The presence of a characteristic melting peak corresponding to the pure drug, without significant shift or disappearance, indicates the absence of major interactions and confirms compatibility with the selected polymers. Conversely, any notable changes in peak position, shape, or enthalpy may suggest potential interactions, polymorphic transformations, or instability within the formulation **.**

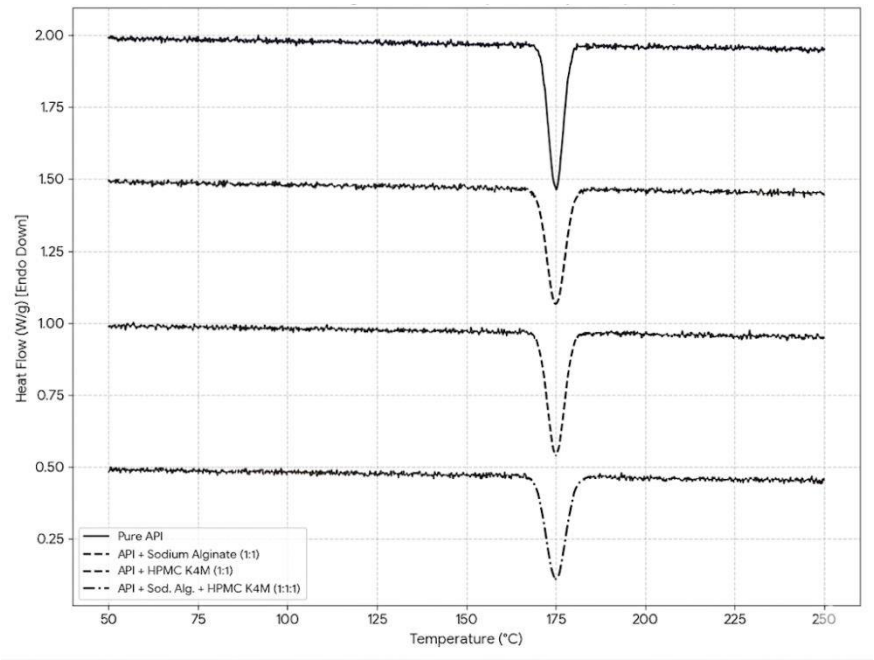


Fig. 23: Drug polymer Compartibility Study of DSC

2. Drug Compatibility Studies: (FTIR)

FTIR analysis confirmed compatibility of drug and polymer. The characteristic peaks for –OH and C–H appeared at 3200–3500 cm^{-1} and 2800–3000 cm^{-1} respectively in the pure API. The fingerprint region is situated at 1600–1000 cm^{-1} the mixtures with Sodium Alginate and HPMC K4M produced the peaks without any significant shift, loss or new peak. Due to physical mixing and possible hydrogen bonding, a minor peak broadening and intensity change occurred. No significant interaction between the API and the polymers was observed, indicating suitability for formulation development.

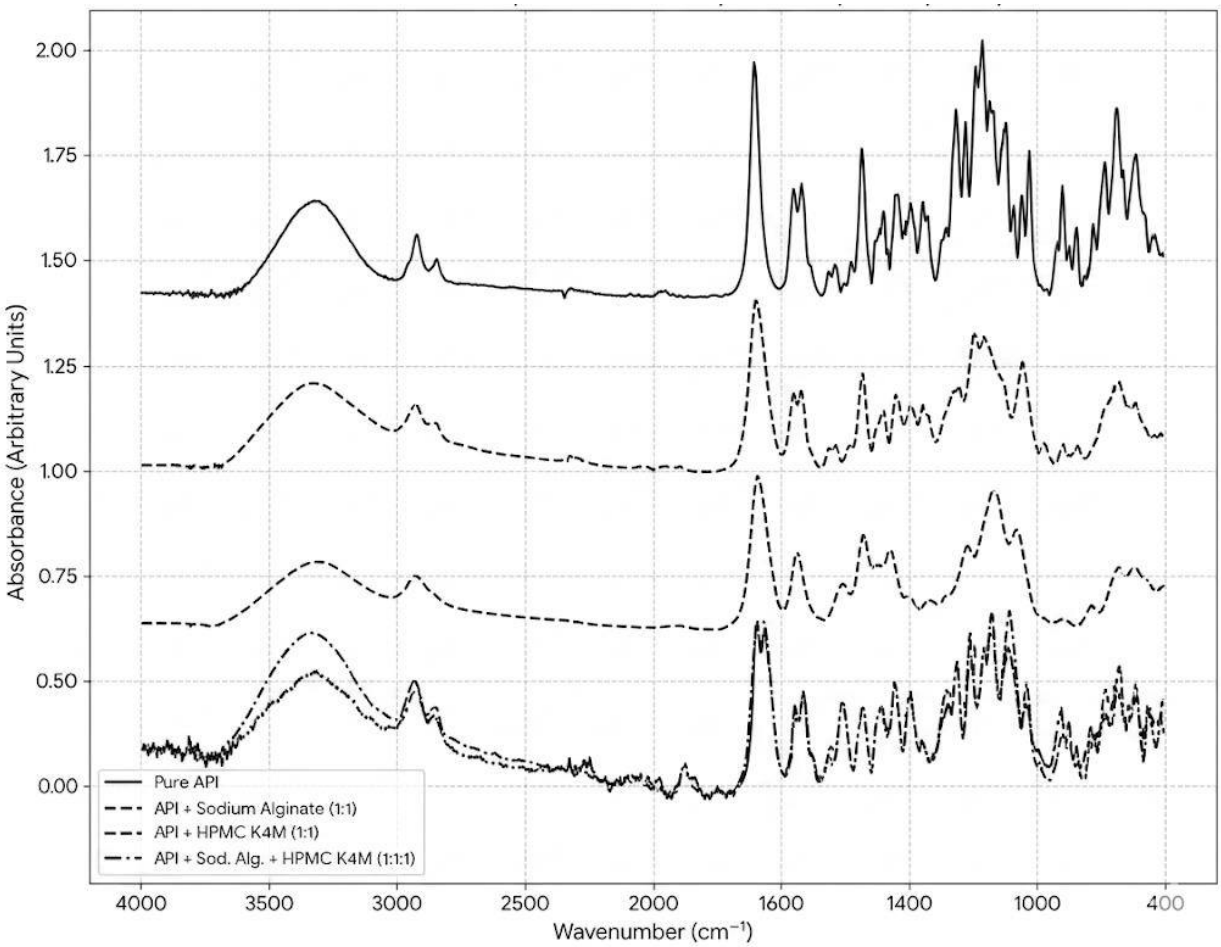


Fig. 24: Drug compatibility study of FTIR

5.3 Evaluation Parameters of Curcumin In-Situ Gel :

Table 7: Evaluation Parameters:

Evaluation Parameters	F1	F2	F3	F4	F5	F6	F7	F8	F9
Gelation Time (sec)	9±0.2	7±0.3	5±0.1	6±0.2	4±0.1	3±0.1	5±0.2	3±0.1	4±0.2
Swelling Index (%)	10.2	12.8	14.9	13.5	15.8	17.4	15.2	17.8	18.6
Gel Strength (N/m^2)	6.8±0.3	7.5±0.4	8.2±0.5	8.0±0.3	9.1±0.6	9.8±0.4	9.3±0.5	10.2±0.3	10.8±0.4
Physical Appearance	Clear	Clear	Clear	Clear	Clear	Clear	Slight viscous	Viscous	Viscous
pH	6.2	6.3	6.4	6.4	6.6	6.7	6.8	6.9	7.0
Density (g/cm^3)	0.55±0.02	0.52±0.01	0.50±0.02	0.50±0.03	0.48±0.01	0.45±0.02	0.46±0.02	0.42±0.01	0.40±0.01
Floating Lag Time (sec)	85±0.5	70±0.4	60±0.3	65±0.3	50±0.2	42±0.2	48±0.3	35±0.2	30±0.2
Duration of Floating (hr)	10±0.2	14±0.3	18±0.4	15±0.2	20±0.5	22±0.4	21±0.3	24±0.5	25±0.6
Drug Content (%)	97.5	98.2	98.8	98.0	99.1	99.4	98.6	99.5	98.9

5.3.1 Determination of Gelation Time:

The floating curcumin formulations (F109) were visually gelated under simulated physiological gastric conditions (0.1 N HCl, pH 1.2 and held at 37 0 C). Specifically, 1 mL of each liquid formulation was gently introduced onto the surface of the unagitated acidic medium to mimic initial gastric entry. When it was in contact with the fluid, calcium ions were released and rapidly cross-linked sodium alginate ions with the solution, inducing an immediate change in state to a liquid (sol) into a highly structured and three-dimensional polymeric gel. The exact time that was taken to fully gel each sample was noted. All the assessments were performed three times to ensure statistical reliability. The difference in the sol-to-gel transition kinetics of the various batches was directly linked to the different ratios of sodium alginate, HPMC K4M, and calcium carbonate that together affect the rate and degree of the polymeric network formation

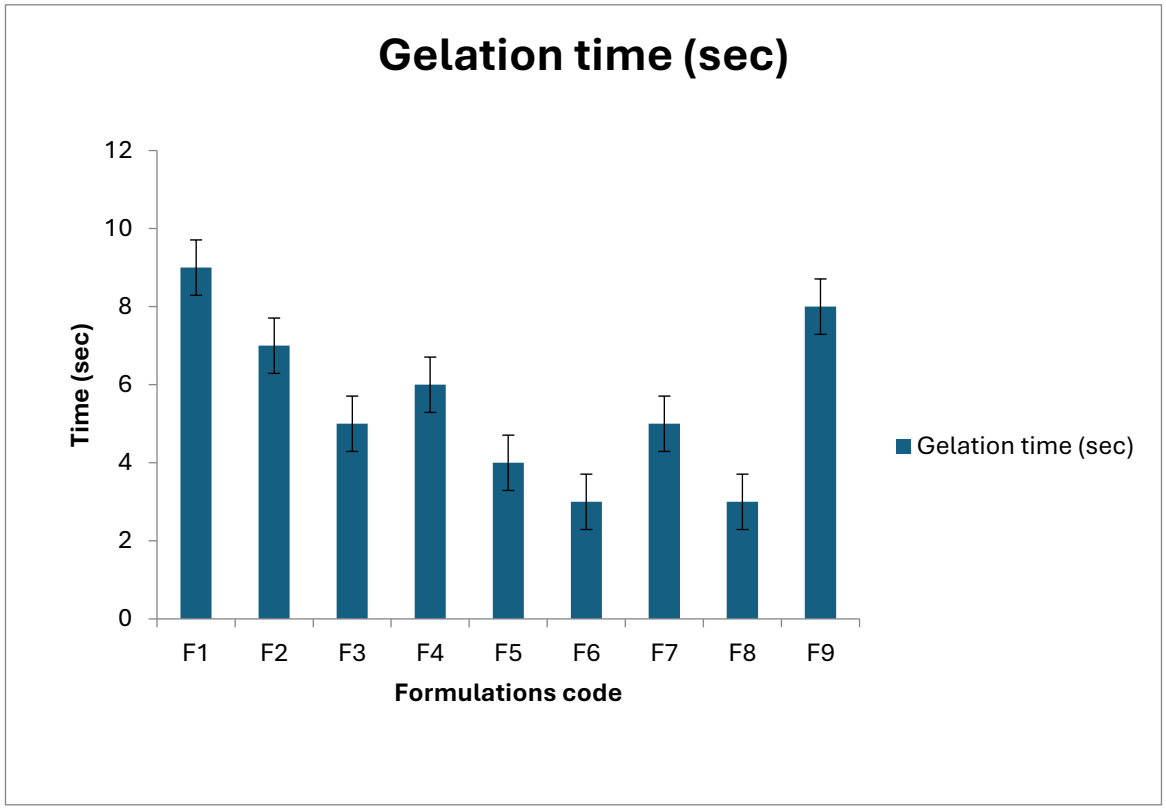


Fig. 19: Graph For Gelation Time

5.3.2. Swelling Index:

To comprehend the dynamics of hydration, matrix expansion and further influence on diffusion of drugs, the swelling capacity of the in-situ gel formulations (F1-F9) was measured. An evident concentration-dependent pattern was visible, where the swelling index was steadily rising between F1 (minimum) and F9 (maximum). This increased absorption of water, especially in batches F5 to F9 are directly caused by the increased levels of hydrophilic polymers in the matrix. This improved and consistent swelling behavior is ultimately crucial in the development of a strong polymeric network, which is fundamental to support long-term gastric floatation, as well as controlled release of curcumin.

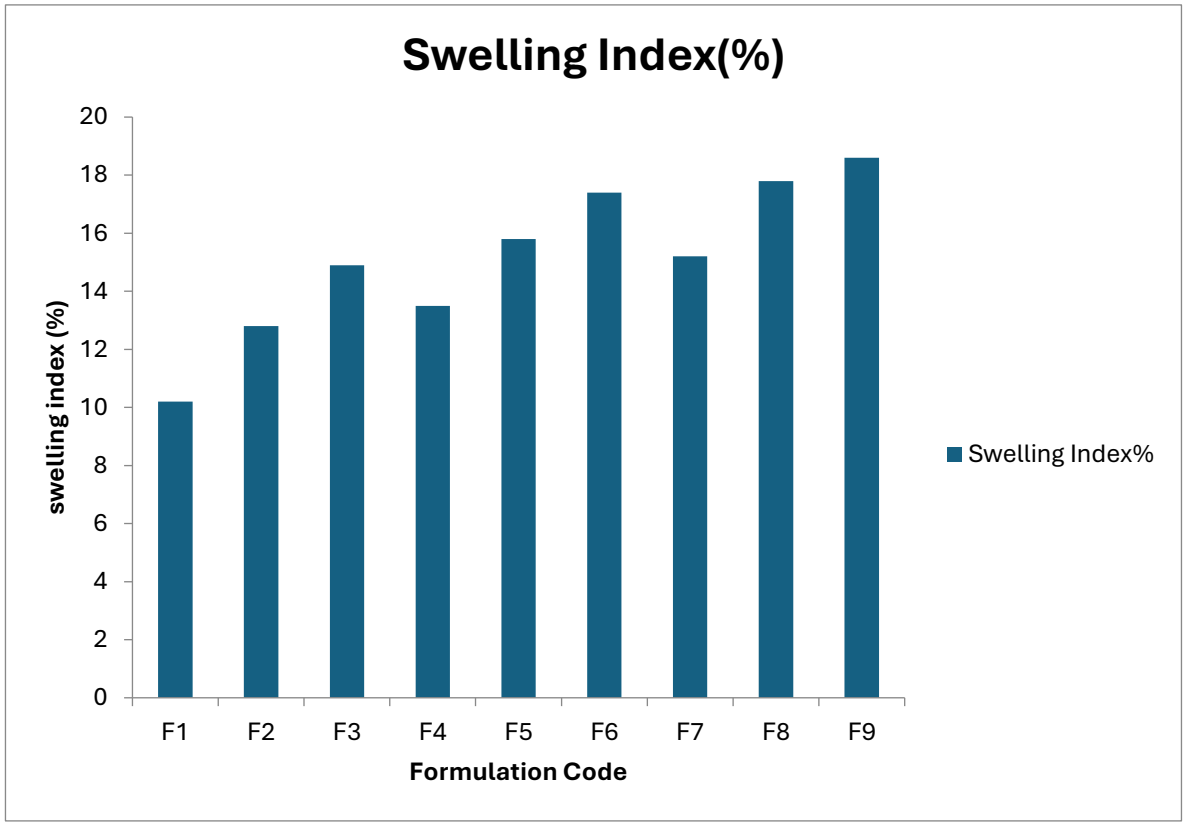


Fig. 20: Graph for Swelling index Percentage Comparison

5.3.3 Gel Strength:

Mechanical integrity of in situ gel formulations (F1-F9) was tested to determine their structural integrity and deformation resistance, which is essential to provide a controlled drug release profile. The analysis showed that the strength of the gel increased progressively and concentration-dependently with the batches with a

minimum of 6.8 +/- 0.3 N/m2 (F1) and a maximum of 10.8 +/- 0.4 N/m2 (F9) strengths. This rising tendency is directly related to rising levels of the polymeric ingredients in the latter formulations, which enable an increased level of ionic cross-linking and consequent creation of a more solid, strong 3D network. Interestingly, formulations F6-F9 showed high mechanical resistance in particular, and this is a sign of high structural durability. These results support the fact that the produced polymeric scaffolds have sufficient gel strength to support the dynamic physiological forces of the gastrointestinal environment, and, thus, provide a stable and reliable platform in long-term therapeutic delivery.

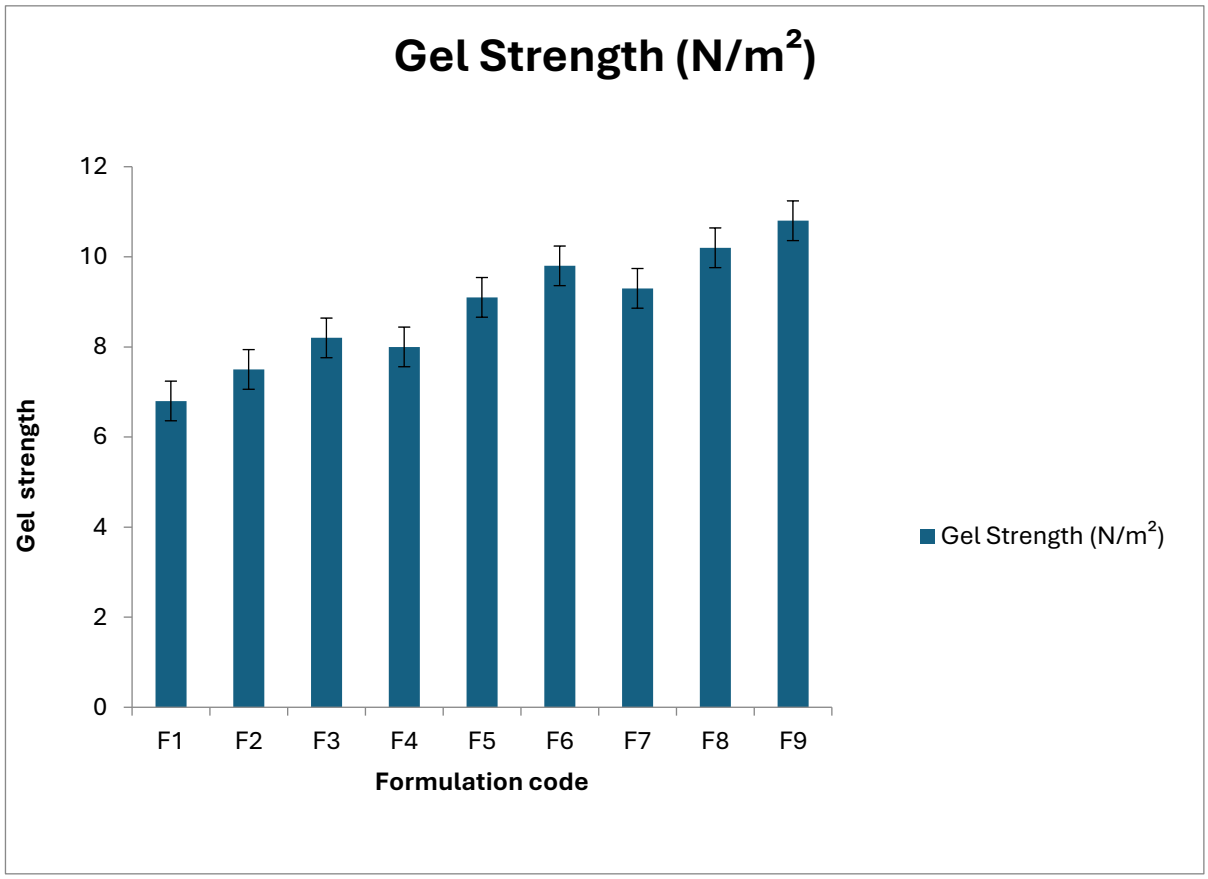


Fig. 21: Graph for Gel strength comparison

5.3.4: Evaluation of Formulation pH

The pH of the in-situ gel formulations (F1-F9) was tested to confirm their compatibility with physiology and stability during pre-administration. The pH values recorded were in a narrow, almost-neutral range of 6.2 (F1) to 7.0 (F9). This slow, gradual movement towards the neutral is explained by the gradual differences in the concentrations of the polymers in the corresponding batches. Importantly, all the measured pH values are within acceptable limits of oral liquid dosage forms. This near-neutral profile ensures the formulations will not be irritating to the gastrointestinal mucosa when ingested and that the formulations will remain in a stable form (sol) before subjected to the acidic gastric environment. Thus, nine formulations were physiologically appropriate and safe to be used as an in-situ gel delivery system.

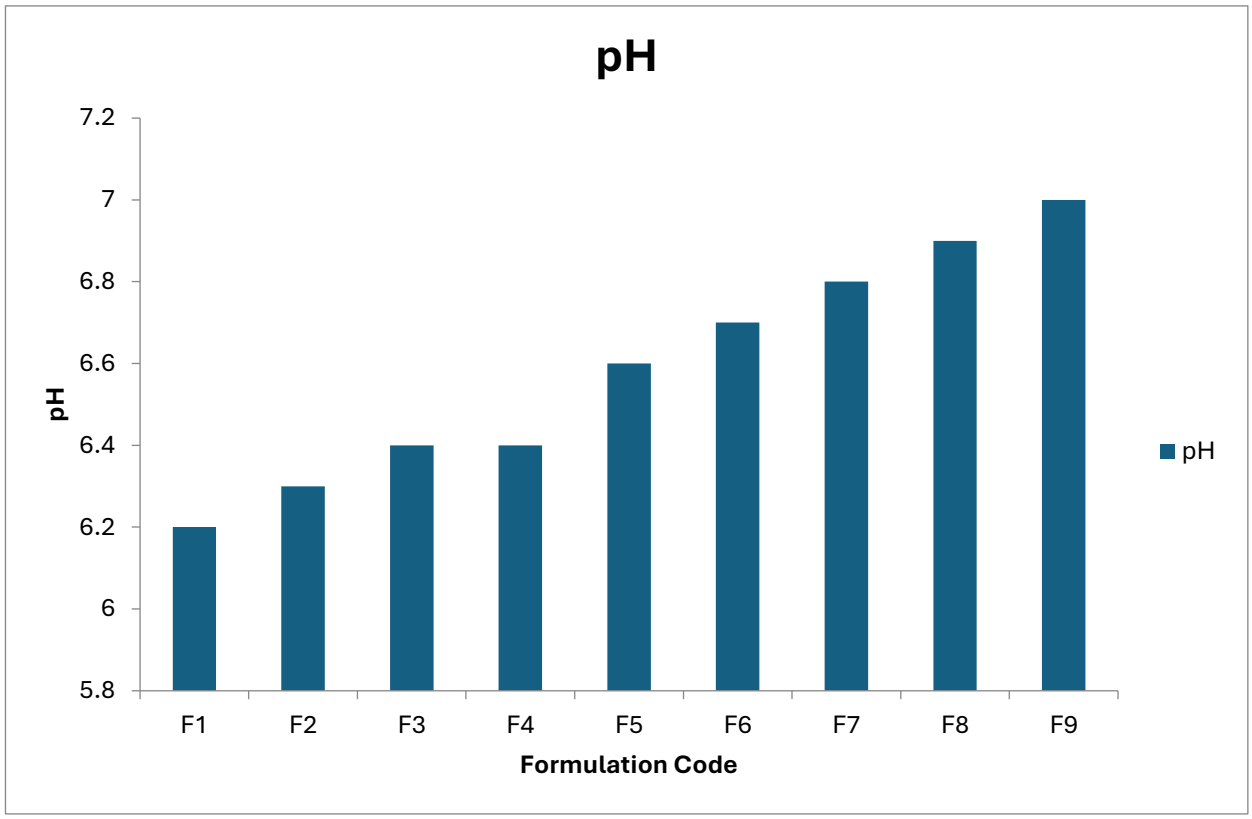


Fig. 22: Graph for PH Comparisons

5.3.5. Density Measurement:

The in-situ floating formulations (F1-F9) were analyzed based on gastroretentive ability (the densities of the in-situ formulations) and showed that the densities decreased gradually, with the highest density of 0.55 ± 0.02

g/cm 3 (F1), and the lowest density of 0.40 ± 0.01 g/cm 3 (The diminishing tendency is mainly attributed to certain changes in polymer concentrations as well as activity of the gas-generating agents incorporated. Above all, the specific gravities of all batches were kept at a significantly low level of specific gravities of physiological gastric fluid (around 1.004 g/cm 3).Such a strong low density property is an essential requirement of floating systems since it ensures the short- and long-term buoyancy of the polymeric matrix, thus confirming the physical adequacy of the formulations to deliver gastroretentive drugs over a long period of time.

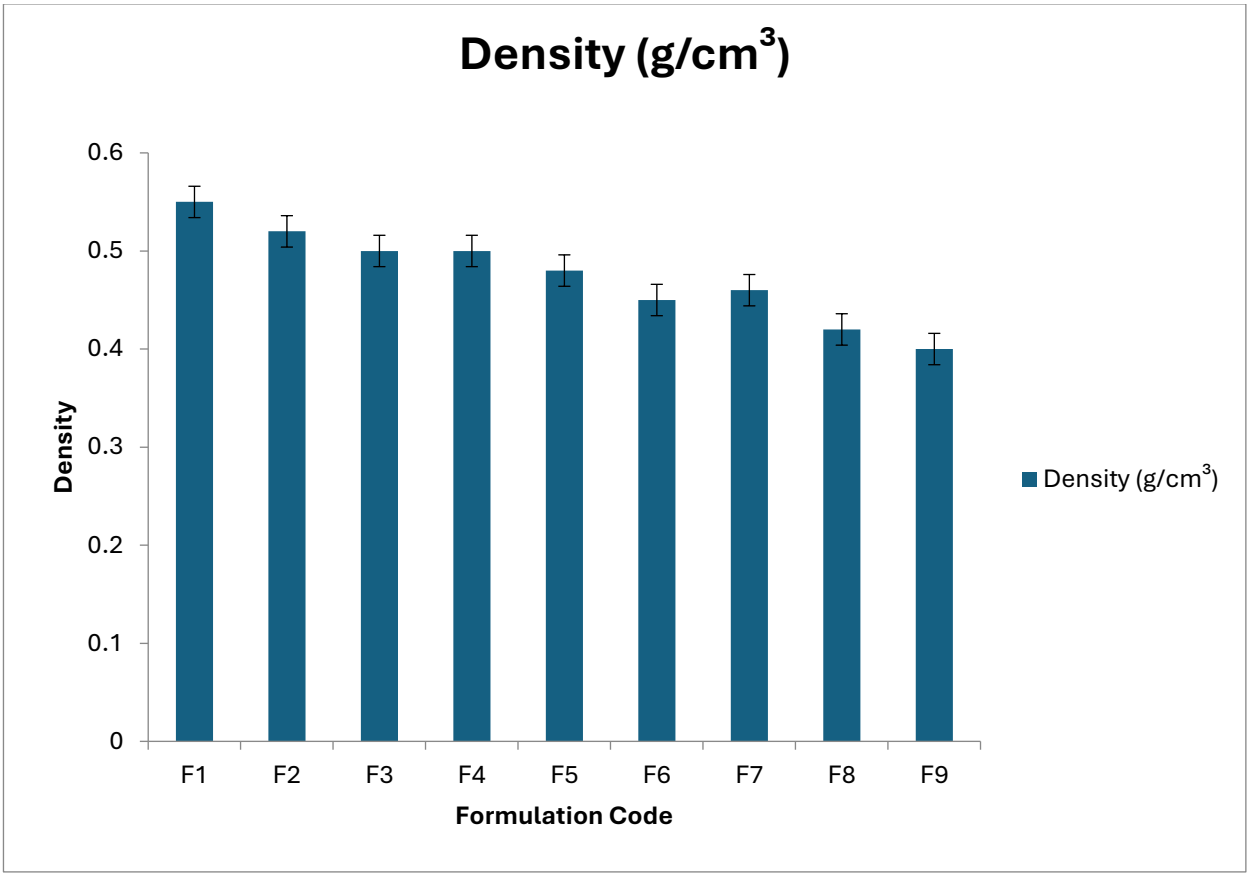


Fig. 23: Graph for Density Comparisons

5.3.6. Floating Lag Time:

The floating lag time of the in-situ gel formulations (F1-F9) was determined to determine the speed of the onset of buoyancy when the formulations were exposed to the simulated gastric fluid. There was a large,

gradual decrease in lag time between the batches, with a high maximum lag time of 85 s + 0.5, with a low minimum of 30 s + 0.2.This enhanced flotation in the later formulations is directly related to the increased concentrations of polymeric and gas generating excipients. The increase in the concentration of these components leads to accelerated sol-gel shifts and faster formation and entrapment of carbon dioxide bubbles in the three-dimensional framework. This, therefore, guarantees a rapid shift to a stable buoyant state, which is a crucial attribute towards the prevention of the formulation experiencing premature gastric emptying.

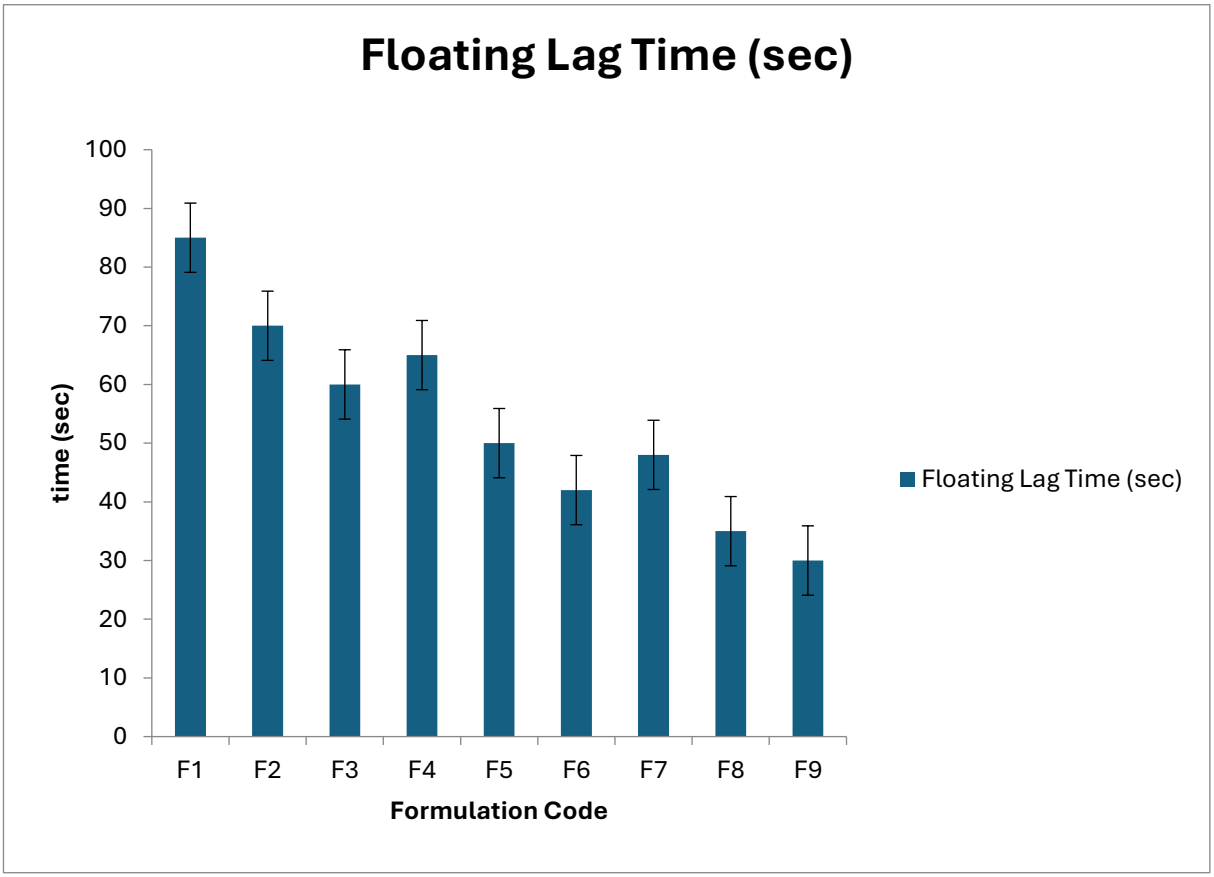


Fig. 24: Graph for Floating Lag time

5.3.7. Total Floating Duration:

The floating time of the in-situ gel formulations (F1-F9) was measured to determine their ability to retain in the gastrointestinal tract. The observed floating times were found to increase progressively, concentration-

dependently across the batches with a minimum of 10 + 0.2 hours (F1) and a maximum of 25 + 0.6 hours (F9).This long buoyancy exhibited in the increased-concentration formulations is directly associated with the enhanced viscosity and enhanced mechanical strength of the gel of the polymeric matrices. A higher density and strength three-dimensional network are better resistant to structural degradation and is also more efficient in retaining the generated carbon dioxide gas. This structural integrity then guarantees that the gel is steadily active in the gastric environment, making the formulations suitable to the sustained and long-term delivery of drugs in the gastric environment.

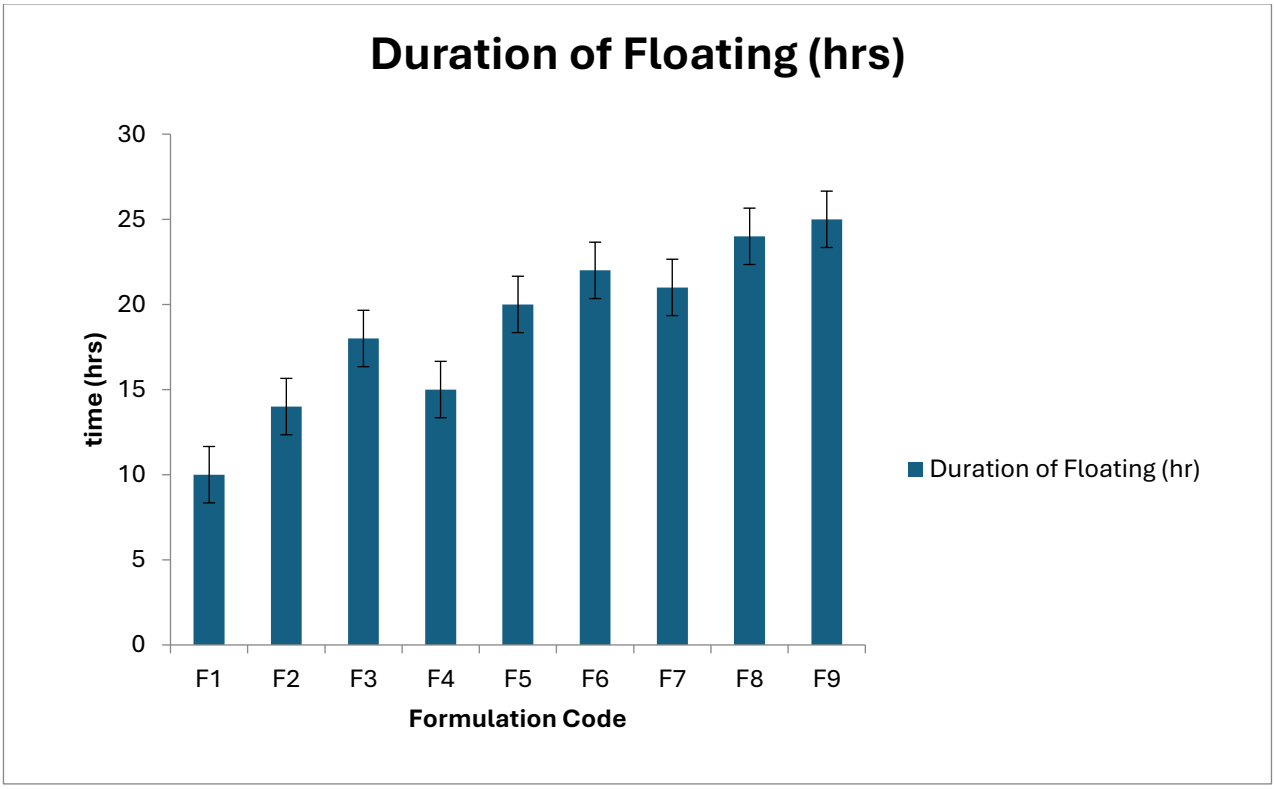


Fig. 25: Graph for Duration of Floating hr Comparison

5.3.8. Drug Content Uniformity:

The drug loading test showed that there was a high consistency in all the batches and the percentage of the curcumin was between 97.5 and 99.5 of the actual percentage. In particular, there was very little variation in

the values of the drug assays, with F1 97.5% and F8 reaching 99.5%. These findings affirm that the preparation method employed is quite stable and guarantees uniformly dispersed active pharmaceutical ingredient within the polymeric sol and consequently, the same and reproducible drug delivery in the varied formulations.

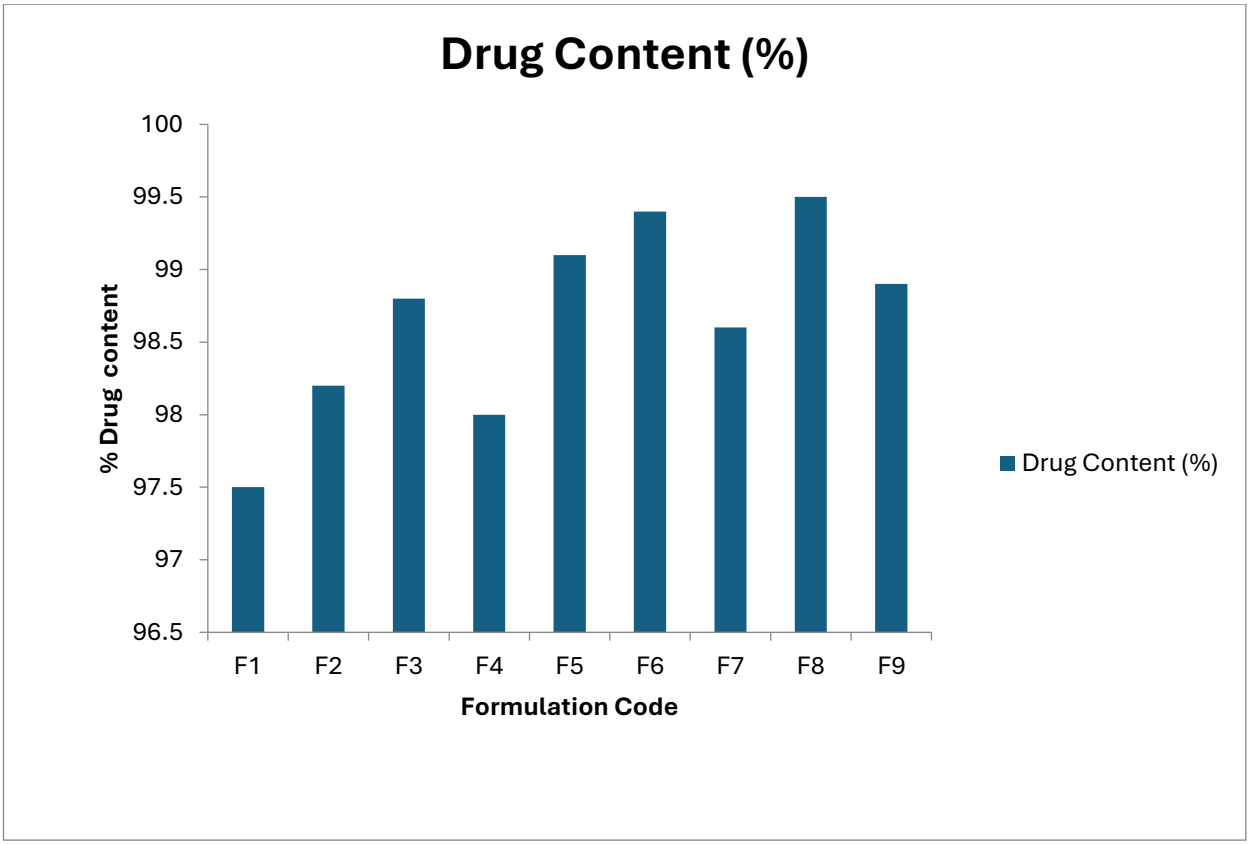


Fig. 26: Graph for Drug Content Percentage

5.4 Impact of Shear Stress on the Viscosity of Formulations:

The rheological characteristics of the in-situ gel formulations (F1-F9) were tested at a constant shear rate between 10-100 rpm. The batches all exhibited typical non-Newtonian, pseudoplastic (shear-thinning) characteristics, with viscosity steadily decreasing with an increase in shear rates. This is achieved since

increased shear rates make the previously entangled polymer chains parallel to the direction of flow, thus it decreases internal fluid resistance.

In addition, the dependence on the concentration was evident: the viscosities of the formulations with higher concentrations of the polymers sodium alginate and HPMC K4M (particularly F7–F9) were significantly higher than those with lower concentrations (F1, F2, F3). The pseudoplastic behavior is most beneficial to oral in-situ delivery systems; it is easy to administer doses by reducing viscosity due to the dynamic shear of pouring and swallowing, but increases the viscosity at rest to allow the development of a strong gel network when the solution reaches the stomach. Interestingly, formulations F8 and F9 had the greatest overall viscosities, which was an indication of their greater ability to create robust, cohesive networks that could be effective in extending gastric retention.

5.4.1 Effect of Shear Stress on Viscosity of Formulations :

Table 8: Effects of Shear Stress

SS (rpm) / Formulation	10	20	30	40	50	60	100
F1	580	530	496	460	410	380	270
F2	650	610	575	540	510	470	350
F3	720	680	640	600	560	520	410
F4	830	790	750	710	670	630	510
F5	950	910	870	830	790	750	620
F6	1080	1040	1005	960	920	880	740
F7	1210	1180	1140	1100	1060	1020	880
F8	1360	1320	1280	1240	1200	1160	1010
F9	1520	1480	1440	1400	1360	1320	1180

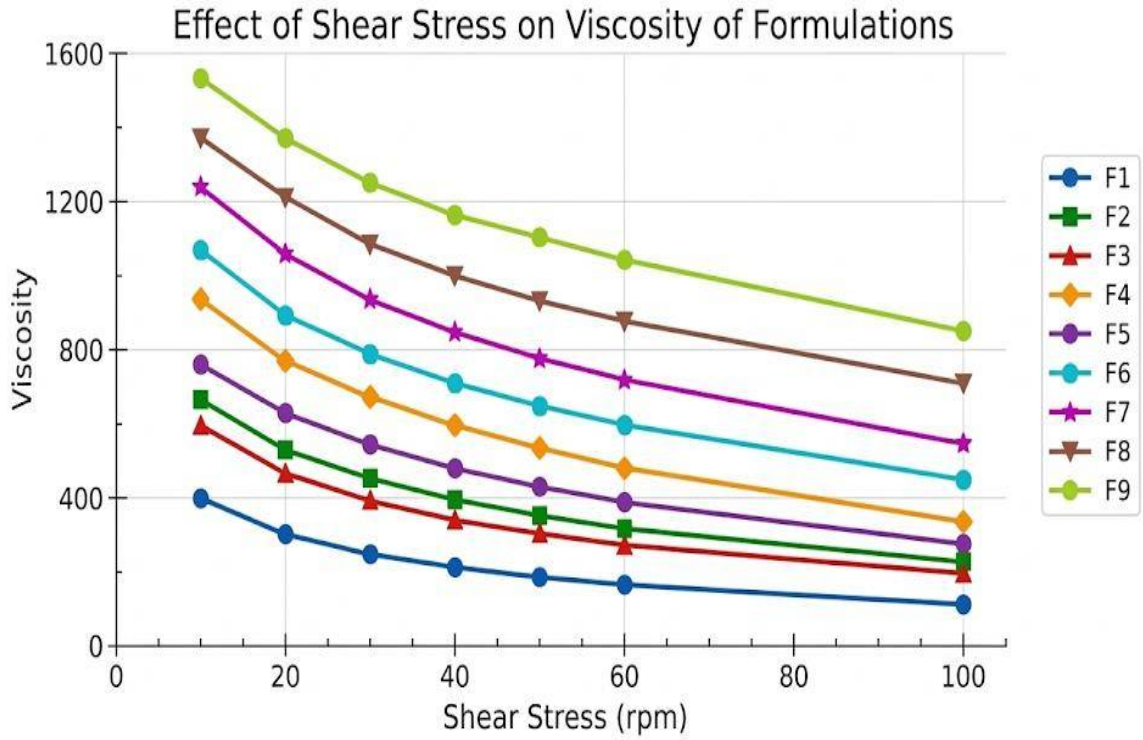


Fig. 27: Effects of Shear Stress

5.5. In-Vitro Drug Release Study:

To assess the physiological performance of in-situ gel formulations of curcumin (F1-F9) floating in-vivo, an in-vivo release study was undertaken in order to measure the physiological performance of the formulations at a 12-hour period. The release profile was biphasic, with a burst effect (first 120-140 minutes) due to the rapid intake of biological fluids and dissolution of the drug on the surface, and a second phase of controlled release (particularly in the final 24 hours) mediated by the restrictive sodium alginate network. On a quantitative basis, the cumulative percentage release decreased with the concentration of all the formulations, with F1 showing the highest rate of 97.84 and F9 showing the lowest rate of 78.36. This gradual decrease in F1 to F9 release rate is directly proportional to the increasing polymer concentrations which form a more compact structural network and also increase the length of the diffusion path of the drug in the gel.

Table 9: In-Vitro Drug Release Profile of Curcumin Floating In-Situ Gel :

Time (hr)	F1 (% ± SD)	F2 (% ± SD)	F3 (% ± SD)	F4 (% ± SD)	F5 (% ± SD)	F6 (% ± SD)	F7 (% ± SD)	F8 (% ± SD)	F9 (% ± SD)
0	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00
1	17.82 ± 0.24	16.41 ± 0.31	15.63 ± 0.27	14.56 ± 0.29	13.72 ± 0.34	12.95 ± 0.25	11.88 ± 0.33	10.97 ± 0.28	9.85 ± 0.30
2	29.76 ± 0.36	28.42 ± 0.40	26.85 ± 0.32	24.94 ± 0.37	23.71 ± 0.41	22.63 ± 0.35	21.24 ± 0.39	19.85 ± 0.33	18.76 ± 0.38
3	41.63 ± 0.42	39.78 ± 0.38	37.94 ± 0.44	35.82 ± 0.40	34.11 ± 0.46	32.76 ± 0.37	31.25 ± 0.43	29.64 ± 0.39	27.92 ± 0.41
4	54.28 ± 0.51	52.17 ± 0.48	49.86 ± 0.46	47.32 ± 0.52	45.68 ± 0.50	43.94 ± 0.47	41.83 ± 0.53	39.92 ± 0.49	37.76 ± 0.55
6	69.94 ± 0.63	67.83 ± 0.60	64.75 ± 0.58	61.47 ± 0.66	58.92 ± 0.61	56.84 ± 0.64	54.26 ± 0.59	51.38 ± 0.67	48.72 ± 0.62
8	82.16 ± 0.72	79.48 ± 0.69	76.12 ± 0.74	72.83 ± 0.70	69.96 ± 0.73	67.28 ± 0.71	64.53 ± 0.75	61.72 ± 0.68	58.94 ± 0.76
10	91.38 ± 0.84	88.64 ± 0.81	85.47 ± 0.79	82.31 ± 0.86	79.56 ± 0.83	76.92 ± 0.80	74.16 ± 0.88	71.45 ± 0.82	68.73 ± 0.87

12	97.84 ± 0.91	95.72 ± 0.88	93.18 ± 0.86	90.46 ± 0.93	88.17 ± 0.89	86.02 ± 0.90	83.54 ± 0.87	80.92 ± 0.94	78.36 ± 0.91

5.6 DRUG RELEASE KINETICS:

5.6.1Zero Order Drug Release Data for Formulations:

The cumulative analysis of the percentage of drug dissolved over time as a function of time was used to plot the zero-order kinetic release profiles of the in-situ gel formulations of curcumin (F1-F9). The release profiles of all experimental batches showed a constant, time-dependent release profile, which confirmed the effectiveness of the polymeric matrix in establishing sustained and controlled release of drugs into the body over 12 hours. It was found that there was a composition-dependent difference in rates of diffusion between the formulations. Formulation F1 had the least dissolution (it was around 65.2 percent), which is explained by extremely strong polymeric network that enhances structural tortuosity and hinders the movement of molecules physically. Conversely, the cumulative drug release rate steadily increased with the successive batches -moderate (F2, F3) to high (F4-F6) release rates; and culminating in essentially 100 percent in the formulations F7, F8, and F9. This enhanced release profile in the later batches is probably due to an augmentation of the matrix porosity and a relative degradation in the localized polymer density that together allows the rapid and unrestricted diffusion of curcumin into the medium surrounding it.

Table 10: Zero-order kinetic Drug release:

Time (hrs)	F1	F2	F3	F4	F5	F6	F7	F8	F9
0	0	0	0	0	0	0	0	0	0
1	8.5	10.2	12.4	15.6	18.1	20.3	22.5	24.2	26.0
2	14.3	17.5	20.8	24.9	28.4	31.6	34.8	37.2	39.5
3	19.7	23.4	27.6	32.1	36.5	40.2	43.9	46.4	49.0
4	24.8	29.2	34.1	39.0	43.6	47.5	51.2	54.0	57.1

5	29.6	34.8	40.2	45.3	50.1	54.0	58.0	61.2	64.5
6	34.2	40.3	46.5	52.0	57.2	61.4	65.6	69.1	72.3
7	39.5	46.1	52.8	58.9	64.3	68.7	72.8	76.5	79.6
8	45.1	52.4	59.8	66.1	71.6	76.2	80.5	84.3	87.1
9	50.4	58.6	66.7	73.0	78.5	83.1	87.2	90.6	93.2
10	55.3	64.5	72.9	79.6	85.0	89.3	92.8	95.6	97.4
11	60.7	70.3	78.8	85.9	90.6	94.2	96.8	98.4	99.2
12	65.2	75.0	84.1	91.5	95.8	98.1	99.2	99.6	100

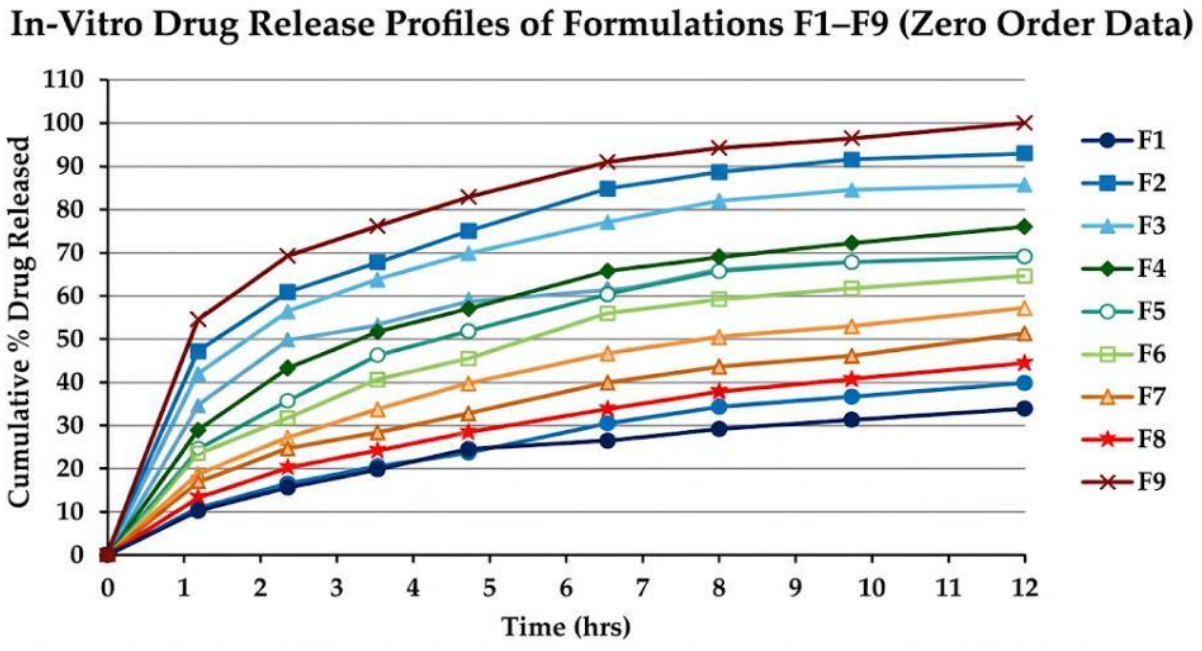


Fig. 28: Zero-Order kinetic

5.6.2 First-order kinetics:

To clarify the dynamics of concentration-dependent release of the in-situ gels (F1-F9), the data obtained during the dissolution were assessed in terms of the first-order kinetic model by plotting the logarithm of the percentage of the unreleased drug vs time. It was found in the analysis that the release rate decreased gradually with time as the active ingredient in the pharmaceutical used up during the 12 hours, a characteristic typical of concentration-sensitive, diffusion-controlled systems. Comparative analysis of regression slopes revealed the existence of recognizably, formulation-dependent, kinetic behaviour's F1-F3 at the onset of a slow, progressive decay, indicative of a highly cross-linked, restrictive polymeric matrix, F4-F6 at the onset of a moderate to intermediate rate of release, and F7-F9 at the onset of a gradual, steep decay, indicative In the end, this first-order mathematical modelling ascertains that the discharge of curcumin out of these delivery systems is rigidly concentration-related as well as is essentially controlled by the physical density and the particular composition of the developed polymeric network.

Table 11: First-Order kinetics:

Time (hrs)	F1	F2	F3	F4	F5	F6	F7	F8	F9
0	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000
1	1.961	1.952	1.942	1.926	1.913	1.901	1.889	1.879	1.869
2	1.933	1.917	1.899	1.875	1.856	1.835	1.814	1.797	1.781
3	1.904	1.883	1.861	1.832	1.803	1.777	1.749	1.726	1.707
4	1.876	1.850	1.818	1.785	1.751	1.720	1.688	1.662	1.632
5	1.848	1.813	1.777	1.737	1.699	1.662	1.623	1.589	1.551

6	1.818	1.776	1.728	1.681	1.632	1.586	1.536	1.490	1.442
7	1.782	1.731	1.674	1.614	1.552	1.495	1.434	1.371	1.309
8	1.740	1.678	1.603	1.530	1.453	1.375	1.289	1.198	1.110
9	1.695	1.616	1.523	1.431	1.332	1.229	1.108	0.974	0.833
10	1.650	1.550	1.432	1.309	1.176	1.029	0.857	0.643	0.415
11	1.594	1.473	1.325	1.146	0.972	0.763	0.505	0.204	0.079
12	1.541	1.398	1.201	0.929	0.623	0.278	0.079	0.018	0.000

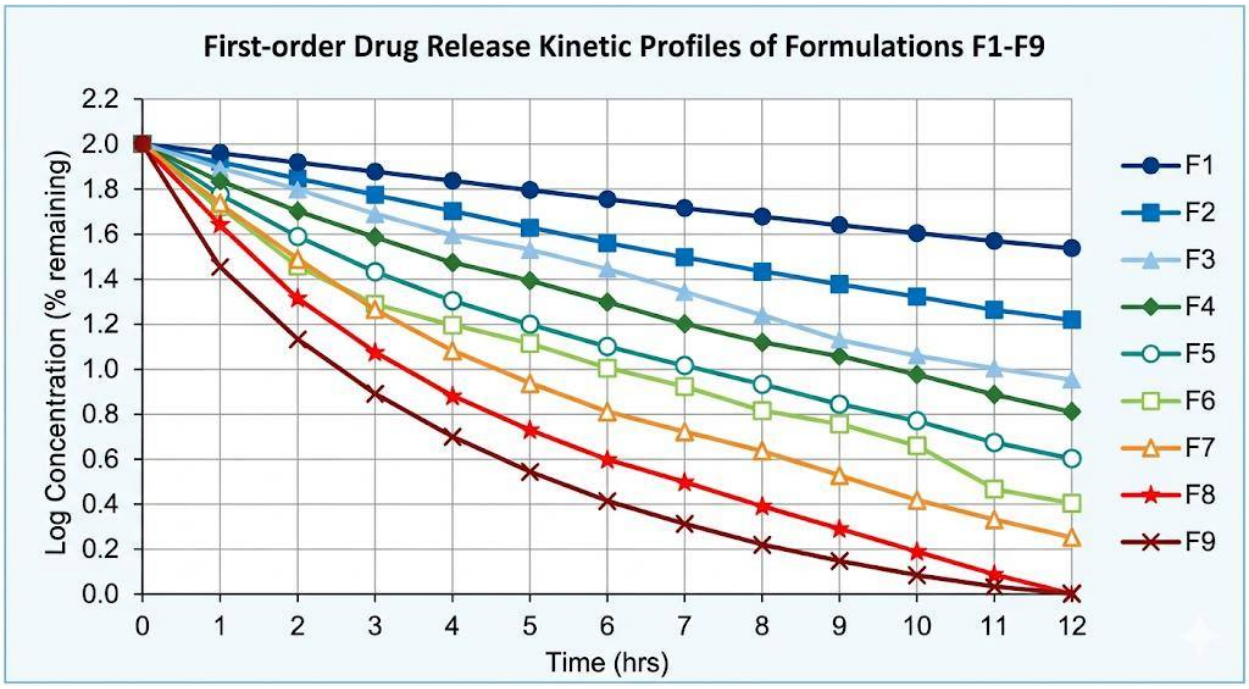


Fig. 29: First-Order kinetics

5.6.3 Higuchi Release Model:

To explain physical release mechanism of the drug out of the in-situ gel systems (F1-F9), the data of dissolution were analyzed through Higuchi kinetic model as a plot of the percentage of drug released

cumulatively with regard to time (√t). A very linear and proportional relationship was observed in all experimental batches, which conclusively confirmed that molecular diffusion through the insoluble polymeric matrix forms the major mechanism that will control the drug release. In line with the past kinetic observations, a structural-dependent diffusion spectrum was revealed: the formulations F1 to F3 were characterized by the slowest release rate, which was reminiscent of a dense, rigid network that severely limits the molecular movement; F4 to F6 realized a moderate and balanced diffusion profile, and F7 to F9 allowed the maximum rate of drug migration. In the end, the high compliance with the Higuchi equation confirms the fact that the sustained release of curcumin is obtained through the sustained and constant process of diffusion of the drug molecules through the matrix pores and into the physiological medium.

Table 12: Higuchi Release Model:

Time (hr)	√Time	F1	F2	F3	F4	F5	F6	F7	F8	F9
0	0.00	0	0	0	0	0	0	0	0	0
1	1.00	8.5	10.2	12.4	15.6	18.1	20.3	22.5	24.2	26.0
2	1.41	14.3	17.5	20.8	24.9	28.4	31.6	34.8	37.2	39.5
3	1.73	19.7	23.4	27.6	32.1	36.5	40.2	43.9	46.4	49.0
4	2.00	24.8	29.2	34.1	39.0	43.6	47.5	51.2	54.0	57.1
5	2.24	29.6	34.8	40.2	45.3	50.1	54.0	58.0	61.2	64.5
6	2.45	34.2	40.3	46.5	52.0	57.2	61.4	65.6	69.1	72.3
7	2.65	39.5	46.1	52.8	58.9	64.3	68.7	72.8	76.5	79.6
8	2.83	45.1	52.4	59.8	66.1	71.6	76.2	80.5	84.3	87.1
9	3.00	50.4	58.6	66.7	73.0	78.5	83.1	87.2	90.6	93.2
10	3.16	55.3	64.5	72.9	79.6	85.0	89.3	92.8	95.6	97.4
11	3.32	60.7	70.3	78.8	85.9	90.6	94.2	96.8	98.4	99.2

12	3.46	65.2	75.0	84.1	91.5	95.8	98.1	99.2	99.6	100

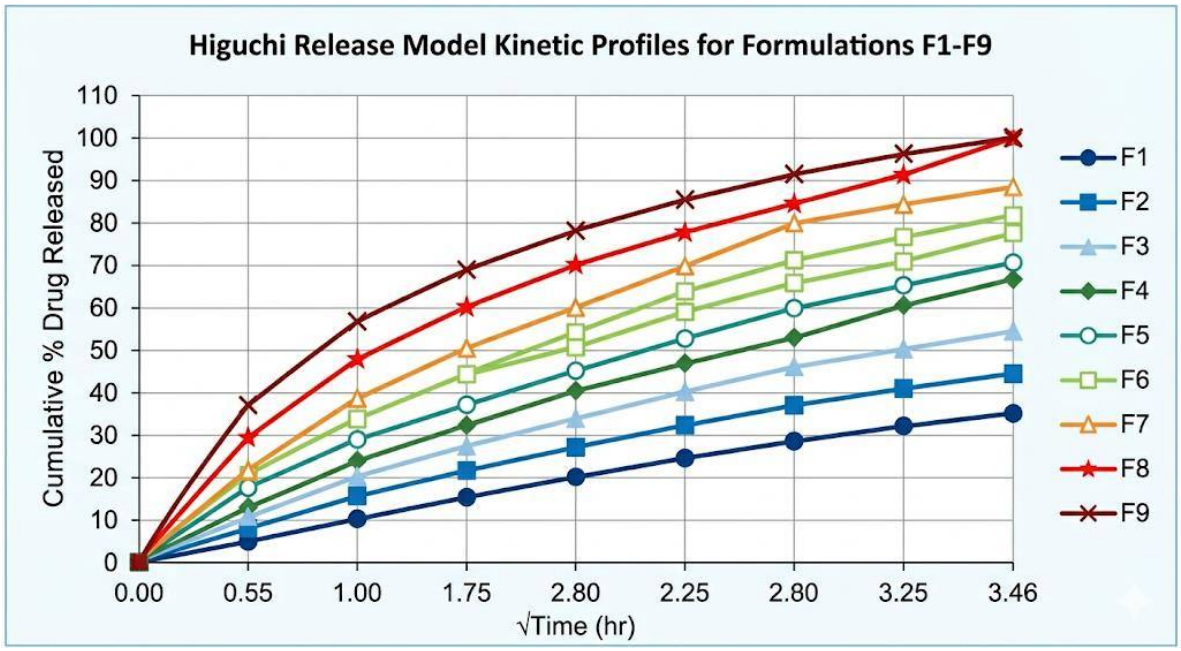


Fig. 30: Higuchi Release Model Kinetics

5.6.4 Korsmeyer–Peppas Release Model:

As a further explanation to the exact processes controlling curcumin discharge, the data were analyzed with the Korsmeyer-Peppas mathematical model plotted with an inverse of the logarithm of time verses the inverse of the logarithm of the percentage of drug released. Experimental formulations have shown that all the formulations have strong linear correlations and the release profiles followed are strictly sustained by Peppas kinetics. It was again found that there is a composition-dependent release gradient: release rates were the slowest with formulations F1F3, and fastest with formulations F4F6F7F9, which indicates that in F1F3 the polymeric matrix is highly cross-linked and creates a strictly diffusion-restrictive barrier, in F4F6F7F9, the relatively slower rate is due to the complex release process of a combination of Most importantly, there is a strong linearity of these plots which enables the determination of the diffusional release exponent (n) with the slope of the regression. This exponent is critical in describing the prevailing mode of drug transport, allowing one to consider the behavior of the system as being either standard Fickian diffusion ($n < 0.45$), anomalous or non-Fickian transport ($0.45 < n < 0.89$), or Case-II polymer relaxation ($n > 0.89$).

Table 13: Korsmeyer–Peppas Release Model:

Time (hr)	log Time	F1	F2	F3	F4	F5	F6	F7	F8	F9
1	0.000	0.929	1.009	1.093	1.193	1.258	1.307	1.352	1.384	1.415
2	0.301	1.155	1.243	1.318	1.396	1.453	1.499	1.541	1.571	1.597
3	0.477	1.294	1.369	1.441	1.507	1.562	1.604	1.642	1.667	1.690
4	0.602	1.394	1.465	1.533	1.591	1.639	1.676	1.709	1.732	1.756
5	0.699	1.471	1.541	1.604	1.656	1.700	1.732	1.763	1.787	1.809
6	0.778	1.534	1.605	1.667	1.716	1.758	1.788	1.817	1.840	1.859
7	0.845	1.597	1.664	1.723	1.770	1.808	1.837	1.862	1.884	1.901
8	0.903	1.654	1.719	1.776	1.820	1.855	1.882	1.906	1.926	1.940
9	0.954	1.702	1.768	1.824	1.863	1.895	1.920	1.940	1.957	1.970
10	1.000	1.743	1.810	1.863	1.901	1.929	1.951	1.968	1.980	1.989
11	1.041	1.783	1.847	1.897	1.934	1.957	1.974	1.986	1.993	1.996
12	1.079	1.814	1.875	1.925	1.961	1.981	1.992	1.996	1.998	2.000

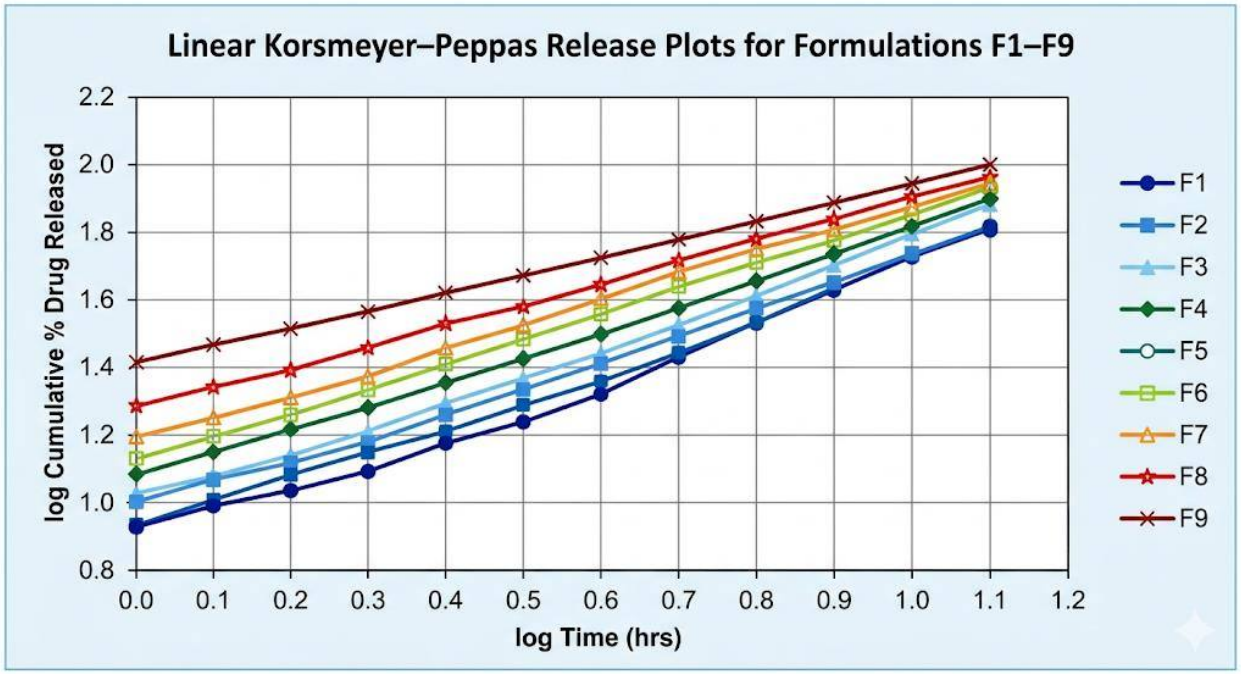


Fig. 31: Korsmeyer-Peppas release mechanism

Conclusion

We designed a floating in-situ gelling system using sodium alginate and HPMC K4M to overcome curcumin's limited solubility in water and low oral bioavailability. After testing the absolute compatibility of the drug with its excipient, we optimized the formulations to become quickly sol-to-gel in the simulated stomach. These gastroretentive gels were highly stable, had uniform drug content, and remained buoyant, although higher quantities of polymer produced more rheological and sustained release properties. Our experimentally evaluated and measured the duration of the gastrointestinal system up to 12 hours in our model that mimicked Higuchi and Korsmeyer-Peppas (non-Fickian) models. The enhanced delivery system significantly prolongs

the length of the gastric and maximized the localized release of curcumin, thus opening up an effective therapy for gastric disorders.

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